AOFC Exercise 06

25.11.2025

Why is the unit of Dispersion ps/(nm km)

ps: in Gigabit and Terabit Systems, ps's are the measurement (range)

nm: laser bandwidth, operation wavelength

km: fiber length

What's the temporal broadening of this pulse inside a fiber




Fabry-Perot-Laser, 0.1 nm spectral width

$$\Delta t_{pulse} = 17\,\frac{ps}{nm \cdot km} \cdot 0.1\,nm \cdot 80\,km = \underline{136\,ps}$$

10 Gbit, OOK → on, off keying → one bit per Symbol

$\frac{1}{R_s} = 100\,ps$ → pulses are overlapping in time, System is limited by Dispersion!

Exercise 10

a) complete Dispersion of the fiber

$$\tau = \underbrace{\frac{n_1 - n_2}{c} \cdot \frac{d(VB)}{dV}}_{\text{waveguide}} + \underbrace{\frac{1}{c} \cdot N_2}_{\text{Material}} \rightarrow \text{derative of } \tau \text{ over } \lambda$$

$$\underbrace{-\frac{n_1 - n_2}{c \cdot \lambda} \cdot V \frac{d^2(VB)}{dV^2}}_{\text{waveguide Dispersion}} - \frac{\lambda}{c} \cdot \frac{d^2 n_2}{d\lambda}$$

$a = 4\,\mu m$ $\quad A_N = 0.14$ $\quad \lambda = 1{,}55\,\mu m$ $\quad D_M = 17\,\frac{ps}{nm \cdot km}$

$$V = k_0 \cdot a \cdot A_N = \frac{2\pi}{1.55\,\mu m} \cdot 4\,\mu m \cdot 0.14 = 2.27 \quad \Rightarrow V\frac{d^2(Vb)}{dV^2} = 0.27$$

$$NA^2 = n_1^2 - n_2^2$$

$$n_1 = \sqrt{NA^2 + n_2^2} = \sqrt{0.14^2 + 1.44^2} = 1.447$$

$$-\frac{n_1 - n_2}{c \cdot \lambda} \cdot V\frac{d^2(Vb)}{dV^2} - \frac{\lambda}{c}\frac{d^2 n_2}{d\lambda}$$

$$-\frac{(1.47 - 1.44)}{3e8\,\frac{m}{s} \cdot 1.55\,10^{-6}\,m} \cdot 0{,}27 + 17\,\frac{ps}{nm\,km} = 12.935\,\frac{ps}{nm\,km}$$

b) $\quad D \stackrel{!}{=} 0$

$$0 = D_W + D_M \qquad D_W = -17\,ps/(nm \cdot km)$$

$$D_W = -\frac{(n_1 - n_2)}{c \cdot \lambda} \cdot V\frac{d^2(Vb)}{dV^2}$$

→ solve for this part

$$V\frac{d^2(Vb)}{dV^2} = D_W \cdot \left(-\frac{c \cdot \lambda}{(n_1 - n_2)}\right)$$

$$= -17\,\frac{ps}{nm\,km} \cdot \left(-\frac{3e8\,\frac{m}{s} \cdot 1.55\,\mu m}{1.447 - 1.44}\right) = 1.129$$

↓ plug into Dispersion Diagramm

$V = 0{,}8$, other solution makes no sense, no mode propagating!

↳ $V = 1.45$

$$a = \frac{V}{k_0 \cdot A_N} = \frac{1.45}{\frac{2\pi}{1.55\,\mu m} \cdot 0.14} = 2.56\,\mu m$$

$a = 2{,}56\,\mu m$ is quite small, complicated fiber coupling!

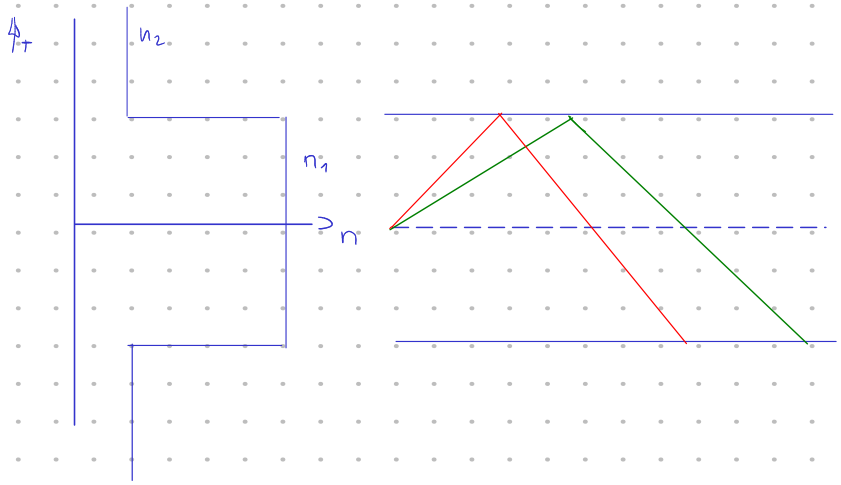
Step Index Fibers

refr. index

$$n(r) = \begin{cases} n_1, & r \le a \\ n_2, & r > a \end{cases}$$

field $E_x(r,\varphi) = E_0 \begin{cases} A_1 \cdot J_\ell\left(r\frac{U}{a}\right) \cdot \begin{cases} \cos(\ell\cdot\varphi) \\ \sin(\ell\cdot\varphi) \end{cases} \\ A_2 \cdot K_\ell\left(r\frac{U}{a}\right) \begin{cases} \cos(\ell\cdot\varphi) \\ \sin(\ell\cdot\varphi) \end{cases} \end{cases}$

number of modes $M = \frac{V^2}{2}$



gradient index Fibers (parabolic shape)

$$n(r) = \begin{cases} n_1^2 \cdot 1 - 2\Delta\left(\frac{r}{a}\right)^2, & r \le a \\ n_1^2 \cdot (1-2\Delta), & r > a \end{cases} \qquad \Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1}$$

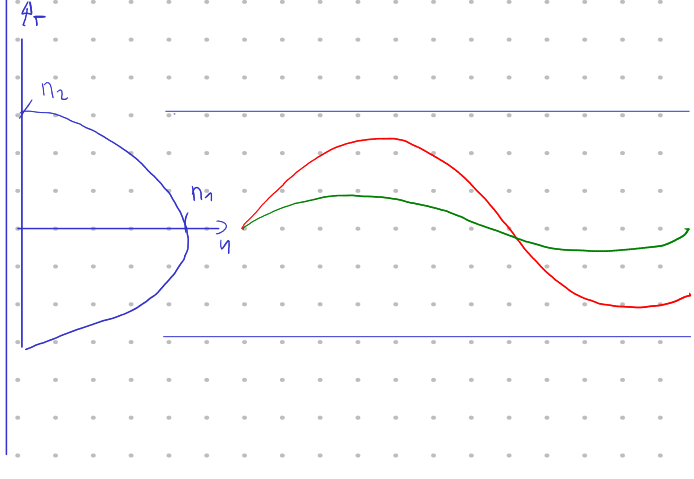
$$E_{x,\ell,p}(r,\varphi) = C'_{LP}\left(\frac{r}{\xi}\right)^{\ell}{}_{L_{P-1}}\left(\frac{r^2}{\xi}\right) e^{\left(\frac{-r^2}{2\xi^2}\right)} \begin{cases} \sin(\ell\cdot\varphi) \\ \cos(\ell\cdot\varphi) \end{cases}$$

$$\beta_{LP} = n_1 \cdot k_0 \left(1 - \frac{\sqrt{2\Delta}}{n_1 \cdot k_0 \cdot a}\right)$$

$(2\ell + 4p - 2)^{1/2}$

$M = \frac{V^2}{4}$

Important for us!



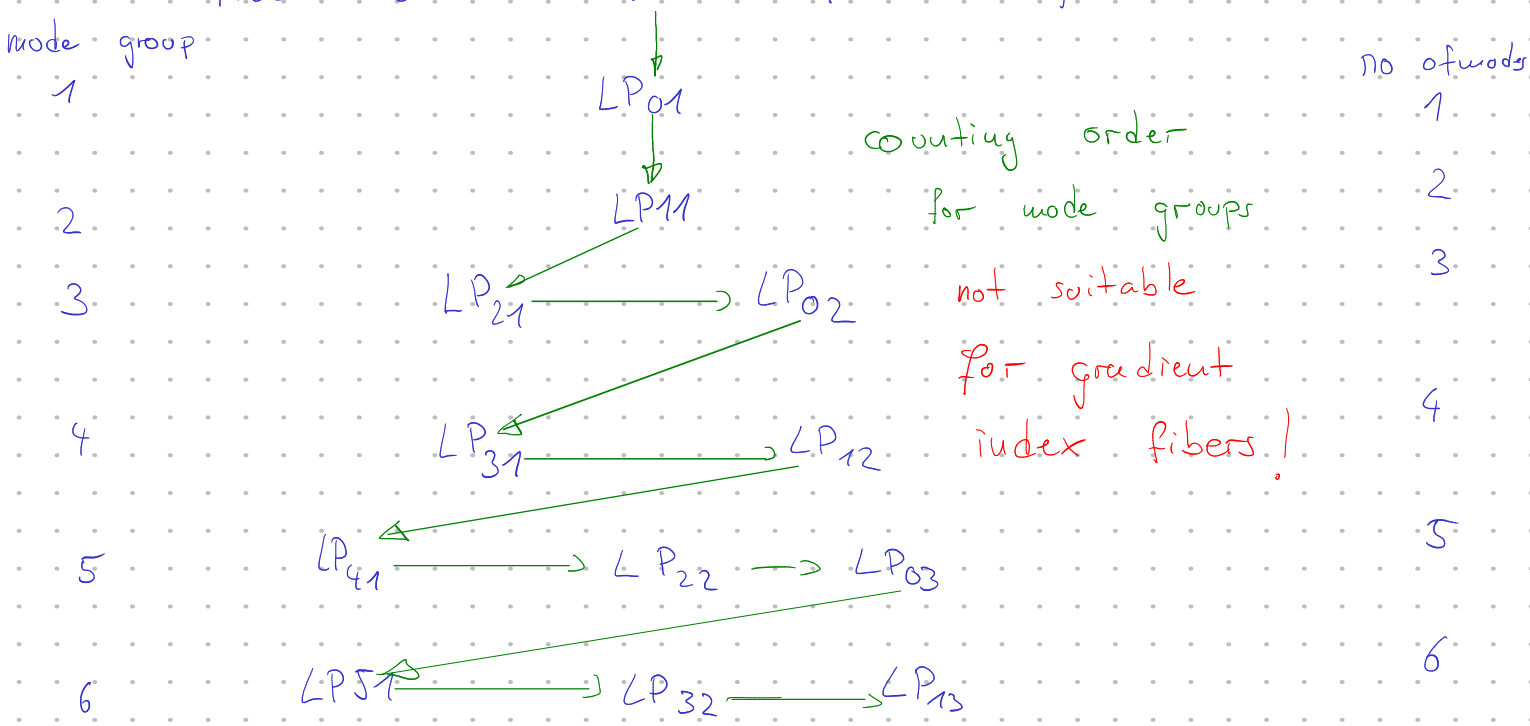
a) supported modes $V = 8$ LP_{01}, LP_{11}, LP_{21}, LP_{02}, LP_{31}, LP_{12}, LP_{41}, LP_{22}, LP_{03}, LP_{51}

b)	LP01	LP11 ×2	LP21 ×2	LP02	LP31 ×2	LP12 ×2	LP41 ×2
$2\ell+4p$	4	6	8	8	10	10	12
MG	1	2	3		4		5

	LP22 ×2	LP03	LP51 ×2
$2\ell+4p$	12	12	14
			6

only mode group 1-3 can propagate inside a gradient index fiber

Mode order for step index fibers



c) $V = 11$ $\quad n_1 = 1.45$, $n_2 = 1.44$

$$\Delta t_\omega = t_{\omega,LP} - t_{\omega,01} = L \cdot \frac{n_1 - n_2}{c_0} \left.\frac{d\,VB}{dV}\right|_{LP} - L \cdot \frac{n_1 - n_2}{c_0} \left.\frac{d\,VB}{dV}\right|_{01}$$

$$= L \cdot \frac{n_1 - n_2}{c_0} \left(\left.\frac{d\,VB}{dV}\right|_{LP} - \left.\frac{d\,VB}{dV}\right|_{01} \right)$$

Step Index Fiber:

$$\left.\frac{d\,(VB)}{dV}\right|_{12} - \left.\frac{d\,(VB)}{dV}\right|_{01} = 0.22 \qquad \Delta t_\omega = L \frac{n_1 - n_2}{c_0} \cdot 0.22$$

$$\Delta t_\omega = 5\,km \cdot \frac{n_1 - n_2}{c_0} \cdot 0.22 = 26.7\,ns$$

Gradient Index Fiber

$$\left.\frac{d\,(VB)}{dV}\right|_{12} - \left.\frac{d\,(VB)}{dV}\right|_{01} = 1.3 \cdot 10^{-3} \qquad \Delta t_\omega = 5\,km \, \frac{1.45 - 1.44}{3e8\,\frac{m}{s}} \cdot 1.3\,10^{-3} = 216.7\,ps$$

$$T_s = \frac{1}{R_s} \doteq 2 \cdot \Delta t_\omega \qquad R_s = \frac{1}{2 \cdot \Delta t_\omega}$$

$$R_{s,Si} = \frac{1}{2 \cdot 26.7\,ns} = 13.62\,MBd$$

$R_{s,GI} \gg R_{s,Si}$

$$R_{s,GI} = \frac{1}{2 \cdot 216.7\,ps} = 2.3\,GBd$$